

PPHA 34600: Program Evaluation
SYLLABUS

Spring 2019: Tuesdays and Thursdays, 9:30-10:50a and 11:00-12:20p, Keller Center 1022.
Course website: Canvas.

Instructor: Prof. Fiona Burlig
Office: Keller Center 2055
Office hours: Wednesdays, 2-3:30P
Email: burlig@uchicago.edu

TAs:

- *Head TA:* Stephen Stapleton, sstapleton@uchicago.edu
- Marissa Block, blockml@uchicago.edu
- Ken Chen, cken@uchicago.edu
- Claire Herdeman, cherdeman@uchicago.edu
- Lindsay Liebert, lindsaypea@uchicago.edu
- Ruodan Liu, ruodanliu@uchicago.edu
- Rohen Shah, shahr@uchicago.edu

TA Sessions and office hours:

- TA sessions will be the first hour, office hours the second hour:
- Mondays, 4-6P, Keller 1002
- Wednesdays, 11A-1P, Keller 2054
- Fridays, 11A-1P, Keller 0010

Course description: The goal of this course is to introduce students to program evaluation, provide an overview of current issues and methods for estimating treatment impacts, and prepare students to be effective consumers of empirical evaluations of real-world policies.

Prerequisites: PPHA31002 and PPHA31102 or equivalent coursework in statistics and economic theory. Students lacking these prerequisites should seek permission from the instructor.

Requirements and grading: Grades will be based on four problem sets and a final exam. Problem sets will count for a total of 75% and the final exam will count for 25%.

Problem sets: Problem sets must be typed and submitted electronically, and late problem sets will not be accepted. Each assignment will receive equal weight. You may work in groups of up to three on your problem sets, ask the course TAs, and get help from Harris' R consultants, but you must turn in your own problem set, with answers written in your own words. All coding in problem sets must be done in R. Due dates are as follows:

- Problem Set 1: Thursday, **April 18** at 9 pm.
- Problem Set 2: Tuesday, **May 07** at 9 pm.
- Problem Set 3: Thursday, **May 16** at 9 pm.
- Problem Set 4: Tuesday, **May 28** at 9pm.



Final exam: The take-home final exam will be assigned on June 2nd, and due at **9 PM on June 5th**. You must do your own work and may not discuss the exam with anyone before it is due. Your exam must be typed and submitted electronically. Late exams will receive a zero.

Re-grade policy: : If you think that there is an error in the grading of your work, you must submit a typed written statement of the details of the problem in question to a TA attached to the assignment in question. The TAs will review both your reasoning and the problem and respond within one week. We reserve the right to re-grade the assignment in its entirety. If you are requesting a re-grade based on other students' grades, you must submit your own problem set and the other student's problem set with your request in order to receive a re-grade. Regrades must be submitted within a week of problem sets being returned.

Readings: Materials for this course consist of two main items: (1) my lecture slides, which will be focused on theory, and (2) a variety of papers that will be available from the course website, which will provide examples of each method. See the schedule and reading list below for topics and associated readings.

Additional policies:

Piazza: The course will have a Piazza site, accessed through Canvas, and maintained by the TAs. If you have content-related questions, please post them on Piazza. Note that the TAs will not respond to Piazza questions submitted within 24 hours of a problem set deadline or the exam deadline.

Email: Please use Piazza over email for questions related to course content. If your non-content-related email cannot be answered in a paragraph or less, I'll ask you to come to office hours or make an appointment so that we can discuss it in person. I will do my best to respond to emails within 48 hours (M-F). If you haven't heard from me within 48 hours, please re-send your email. To greatly increase the likelihood that I (or the TAs) see your email, please be sure to include [PPHA 34600] in the subject line. Neither I nor the TAs will respond to emails sent within 24 hours of a problem set deadline or the exam deadline.

Electronic devices: This course has a no-electronic-devices-in-class policy. Please refrain from using your devices in class. If you require an exemption from this policy (e.g. for a medical reason), please seek permission from the instructor.

Statistical software: Data work for this class, including problem sets and/or the final exam, will be done in R. I recommend that you use RStudio in conjunction with the tidyverse.

Academic honesty: The Harris School has a formal policy on academic honesty that you are expected to adhere to. Examples of academic dishonesty include (but are not limited to) turning in someone else's work as your own, copying solutions to past years' problem sets, and receiving any unapproved assistance on exams. Academic dishonesty will not be tolerated in this course. At a minimum, I will give zeroes on any assignments that include cheating and will strike the highest overall problem set score of any student who has cheated. I will also refer all cases of cheating to the office of the Dean of Students. They may in



turn impose further penalties as per the Harris School Disciplinary Procedures, including as probation and expulsion. If you have any questions regarding what would or would not be considered academic dishonesty in this course, please do not hesitate to ask.

ADA accommodations: Any student who believes they may need assistance should inform the Office of Student Disability Services by the end of the first week of class. Once you have received an accommodation letter, it should be presented to the course instructor immediately. For more information, see <https://disabilities.uchicago.edu/>.

Tentative list of lecture topics and deadlines (subject to adjustments):

- 4/2: Why program evaluation?
- 4/4: Treatment parameters and regression
- 4/9: Randomized controlled trials I
- 4/11: Randomized controlled trials II
- 4/16: Randomized controlled trials III
- 4/18: Evaluation of evaluations
 - **Problem set #1 due**
- 4/23: Selection on observables I
- 4/25: Instrumental variables I
- 4/30: Instrumental variables II
- 5/2: Instrumental variables III
- 5/7: Panel data I
 - **Problem set #2 due**
- 5/9: Panel data II
- 5/14: Panel data III
- 5/16: Regression discontinuity I
 - **Problem set #3 due**
- 5/21: Regression discontinuity II
- 5/23: Big data and machine learning I
- 5/28: Big data and machine learning II
 - **Problem set #4 due**
- 5/30: Does rural electrification work? I
- 6/4: Does rural electrification work? II
 - **Final exam due on 6/5**
- 6/6: Wrapping up

Reading list: Readings will be made available through the course website. I'm not asking you to read much, so all readings are mandatory. Please read the version from the course website to make sure we're all on the same page.

Why program evaluation?

- No readings for the first meeting.



Treatment parameters and regression

- Angrist, Joshua D. and Jorn-Steffen Pischke. 2015. *Mastering 'Metrics*, Princeton University Press: Princeton, NJ: pp 82-97.
- Angrist, Joshua D. and Jorn-Steffen Pischke. 2009. *Mostly Harmless Econometrics*, Princeton University Press: Princeton, NJ: pp 27-64.

Randomized controlled trials I

- Banerjee, Abhijit, Esther Duflo, Rachel Glennerster, and Cynthia Kinnan. 2015. "The miracle of microfinance? Evidence from a randomized evaluation." *American Economic Journal: Applied Economics*, 7(1): 22-53.

Randomized controlled trials II

- Fowlie, Meredith, Catherine Wolfram, C. Anna Spurlock, Annika Todd, Patrick Baylis, and Peter Cappers. 2017. "Default Effects and Follow-on Behavior: Evidence from an Electricity Pricing Program," *NBER Working Paper w23553*.

Randomized controlled trials III

- Bergquist, Lauren, Marshall Burke, and Edward Miguel. 2019. "Sell low and buy high: Arbitrage and local price effects in Kenyan markets," *The Quarterly Journal of Economics*, forthcoming.
- *Optional reading*: Baird, Sarah, J. Aislinn Bohren, Craig McIntosh, and Berk Ozler. 2014. "Designing experiments to measure spillover effects," *IIEP working paper*.

Evaluation of evaluations

- Spurlock, Anna, Peter Cappers, Jing Lin, Annika Todd, and Patrick Baylis. 2016. "Go for the silver? Evidence from field studies quantifying the difference in evaluation results between 'gold standard' randomized controlled trial methods versus quasi-experimental methods," *ACEEE Summer Study on Energy Efficiency in Buildings*, 2-1--2-13.

Selection on observables

- Davis, Lucas W., Alan Fuchs, and Paul Gertler. 2014. "Cash for coolers: Evaluating a large-sale appliance replacement program in Mexico," *American Economic Journal: Economic Policy*, 6(4): 207-238.

Instrumental variables I

- Miguel, Edward, Shanker Satyanath, and Ernest Sergenti. 2014. "Economic shocks and civil conflict: An instrumental variables approach" *Journal of Political Economy*, 112(4):725-753.

Instrumental variables II

- Maccini, Sharon and Dean Yang. 2009. "Under the weather: Health, schooling, and economic consequences of early-life rainfall," *American Economic Review*, 99(3): 1006-1026.

Instrumental variables III

- No new reading. Review Fowlie, Wolfram et al.



Panel data I

- Jensen, Robert. 2007. “The digital divide: Information (technology), market performance, and welfare in the South Indian fisheries sector,” *The Quarterly Journal of Economics*, 122(3): 879-924.

Panel data II

- Hsiang, Solomon M. and Amir S. Jina. 2014. “The causal effect of environmental catastrophe on long-run economic growth: Evidence from 6,700 cyclones,” NBER Working Paper w20352.
- *Optional reading:* Goodman-Bacon, Andrew. 2018. “Difference-in-differences with variation in treatment timing,” *Working paper*.

Panel data III

- Keiser, David A. and Joseph S. Shapiro. 2018. “Consequences of the Clean Water Act and the demand for water quality,” *The Quarterly Journal of Economics*, 134(1): 349-396.

Regression discontinuity I

- Ito, Koichiro. 2015. “Asymmetric incentives in subsidies: Evidence from a large-scale electricity rebate program,” *American Economic Journal: Economic Policy*, 7(3): 209-237.

Regression discontinuity II

- Chen, Yuyu, Avraham Ebenstein, Michael Greenstone, and Hongbin Li. 2013. “Evidence on the impact of sustained exposure to air pollution on life expectancy from China’s Huai River policy,” *Proceedings of the National Academy of Sciences*, 110(32): 12936-12941.

Big data and machine learning I

- Mullainathan, Sendhil and Jann Spiess. 2017. “Machine learning: An applied econometric approach,” *Journal of Economic Perspectives*, 31(2): 87-106.
- Donaldson, Dave and Adam Storeygard. 2016. “The view from above: Applications of satellite data in economics,” *Journal of Economic Perspectives*, 30(4): 171-198.

Big data and machine learning II

- Burlig, Fiona, Christopher Knittel, David Rapson, Mar Reguant, and Catherine Wolfram. 2019. “Machine learning from schools about energy efficiency,” NBER Working Paper w23908.

Does rural electrification work? I

- Dinkelman, Taryn. 2011. “The effects of rural electrification on employment: New evidence from South Africa,” *American Economic Review*, 101(7): 3078-3108.

Does rural electrification work? II

- Burlig, Fiona and Louis Preonas. 2016 “Out of the darkness and into the light? Development effects of rural electrification,” Energy Institute at Haas Working Paper 268.
- Lee, Kenneth, Edward Miguel, and Catherine Wolfram. 2019. “Experimental evidence on the economics of rural electrification,” NBER Working Paper w22292.



- *Optional reading:* Lee, Kenneth, Edward Miguel, and Catherine Wolfram. 2019. “Does electrification supercharge economic development?” *Working paper*.

Wrapping up